

MORRIN AGDM, AB, Canada

WMO: 712570

Lat: **51.6603N**Lon: **112.6750W**Elev: **2743**StdP: **13.30**Time Zone: **-7.00 (NAM)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-22.4	-16.5	-28.7	1.2	-22.3	-22.6	1.7	-16.4	27.0	23.1	24.1	17.7	5.8	190	0.707

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.0	85.6	62.4	81.8	61.2	78.3	59.8	65.8	78.5	63.6	76.9	61.8	74.5	9.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	89.5	71.4	58.7	81.7	69.2	56.6	75.6	66.9	32.1	78.4	30.3	77.3	28.9	74.6	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	22.1	19.2	-30.8	91.7	5.5	3.6	-34.8	94.2	-38.0	96.3	-41.1	98.3	-45.1	100.9
			-30.3	69.8	6.1	2.8	-34.7	71.8	-38.3	73.5	-41.7	75.1	-46.1	77.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.8	13.9	13.5	25.3	39.1	50.5	57.6	62.9	60.7	51.9	39.6	24.2	13.2
	DBStd	21.16	14.81	13.69	14.11	9.34	7.74	5.62	5.38	6.20	8.33	9.28	13.05	13.91
	HDD50	5672	1119	1021	766	339	88	6	0	3	74	340	776	1140
	HDD65	9999	1584	1441	1231	778	450	232	105	160	400	788	1225	1605
	CDD50	1233	0	0	1	11	105	233	399	335	130	18	1	0
	CDD65	82	0	0	0	0	2	9	39	27	5	0	0	0
	CDH74	1874	0	0	1	7	98	193	751	597	217	10	0	0
	CDH80	507	0	0	0	0	11	34	225	176	61	0	0	0
Wind	WSAvg	8.7	8.4	8.1	9.1	10.3	10.4	9.5	7.8	7.8	8.4	8.9	8.4	7.9
Precipitation	PrecAvg	15.1	0.6	0.4	0.6	1.0	1.9	3.0	2.6	2.1	1.3	0.6	0.6	0.5
	PrecMax	19.3	1.3	0.9	2.1	2.4	3.7	6.0	5.9	4.3	3.2	1.4	1.9	1.1
	PrecMin	9.0	0.0	0.0	0.1	0.1	0.2	1.3	0.9	0.1	0.2	0.2	0.1	0.0
	PrecStd	2.6	0.3	0.2	0.4	0.6	0.9	1.2	1.2	1.1	0.7	0.4	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.1	44.0	59.4	73.3	81.2	83.9	90.6	89.5	87.0	74.6	58.1	45.4
		MCWB	36.9	37.0	44.7	51.5	55.5	60.8	63.9	62.4	59.6	53.5	44.3	38.9
	2%	DB	40.6	38.9	52.6	66.9	76.6	79.2	86.0	84.8	80.0	65.9	50.1	39.9
		MCWB	35.2	33.7	41.2	48.1	54.0	59.7	65.1	61.1	57.1	49.9	40.4	35.0
	5%	DB	37.4	35.4	47.3	61.3	72.1	75.3	81.9	80.8	74.8	60.4	44.5	36.1
		MCWB	33.3	31.5	38.6	45.0	52.3	58.0	63.6	61.3	55.8	47.1	37.1	31.8
	10%	DB	33.3	31.9	41.9	56.0	67.7	71.5	77.9	76.5	69.1	55.0	40.3	32.2
		MCWB	30.1	29.0	35.9	42.2	50.1	56.3	62.1	59.8	53.3	44.2	34.9	28.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.8	37.1	46.3	52.1	57.4	64.9	70.3	67.3	60.8	54.5	45.9	39.2
		MCDB	43.4	43.2	57.8	71.8	76.0	78.1	82.4	79.3	81.1	72.6	57.8	45.0
	2%	WB	35.5	34.2	42.0	48.8	55.0	62.0	67.2	64.5	58.6	50.9	41.2	35.2
		MCDB	39.9	38.2	50.9	65.0	72.5	74.9	80.5	78.3	76.8	63.7	49.2	39.8
	5%	WB	33.4	31.7	38.7	45.9	53.3	59.8	64.8	62.5	56.4	47.9	38.0	32.0
		MCDB	37.4	35.2	46.6	59.5	69.6	71.7	78.1	76.3	72.0	59.3	43.4	35.9
	10%	WB	30.3	29.0	35.9	43.1	51.4	57.6	62.9	60.6	54.1	45.1	35.1	28.9
		MCDB	33.3	31.9	41.4	54.3	65.5	68.9	75.3	73.9	68.3	54.2	39.7	32.1
Mean Daily Temperature Range	5% DB	MDBR	17.7	18.6	19.4	23.0	24.6	22.3	26.0	25.8	25.7	23.4	19.1	16.9
		MCDBR	20.1	20.7	25.8	33.6	32.3	29.3	32.8	34.2	36.1	31.8	24.7	19.4
		MCWBR	16.0	16.3	15.9	17.5	14.7	13.3	15.0	14.6	16.3	17.5	16.3	15.3
	5% WB	MCDBR	19.4	20.3	24.6	31.3	29.1	25.2	29.2	29.1	33.3	30.3	23.1	19.3
		MCWBR	15.5	16.4	15.9	16.7	13.5	12.9	14.7	13.8	15.6	16.9	15.8	15.5
Clear-Sky Solar Irradiance	taub	0.266	0.272	0.284	0.338	0.363	0.359	0.356	0.373	0.323	0.283	0.254	0.243	
	taud	2.329	2.404	2.460	2.327	2.310	2.363	2.398	2.333	2.442	2.535	2.484	2.371	
	Ebn at Noon	240	276	294	285	282	284	283	271	275	270	250	234	
	Edh at Noon	19	24	28	36	38	37	35	36	28	21	16	16	
All-Sky Solar Radiation	RadAvg	347	674	1112	1503	1821	1947	2072	1680	1180	711	377	271	
	RadStd	27	55	93	122	119	118	103	119	124	83	32	27	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (81 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page